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GE2 Test cases

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RF1:

Boundary values and equivalence classes:

VARIABLE	RULE	VALID CLASSES	INVALID CLASSES
company_cif	datatype	ECV1 - String	ECNV 1 - Not a string
company_cif	CIF valid	ECV2 - CIF meets validation algorithm	ECNV2 - CIF does not meet validation algorithm (Letter + 7 digits + control)
company_cif	length	ECV3 - CIF has 9 characters	ECNV3 - CIF =! 9
		BVV1 - CIF size 9	BVNV1 - 8 char
			BVNV2 - 10 char
company_cif	content	ECV4 - First char is a letter	ECNV4 - First char is not a letter
project_acronym	datatype	ECV5 - String	ECNV5 - Not a string
project_acronym	size	ECV6 - Between 5 and 10 char	ECNV6 - Acronym < 5
		BVV2 - Acronym size 5	ECNV7 - Acronym > 10
		BVV3 - Acronym size 10	BVNV3 - 4 char
			BVNV4 - 11 char
project_acronym	content	ECV7 - Contains only alphanumeric chars	ECNV8 - Contais non-alphanumeric chars
project_description	datatype	ECV8 - String	ECNV9 - Not a string
project_description	size	ECV9 - Between 10 and 30 chars	ECNV10 - Description < 10
		BVV4 - 10 chars	ECNV11 - Description > 30
		BVV5 - 30 chars	BVNV5 - 9 chars
			BVNV6 - 31 chars
department	datatype	ECV10 - String	ECNV12 - Not a string
department	content	ECV11 - Value "HR"	ECNV13 - Value different from "HR", "FINANCE", "LEGAL", "LOGISTICS"
		ECV12 - Value "FINANCE"	
		ECV13 - Value "LEGAL"	
		ECV14 - Value "LOGISTICS"	
date	datatype	ECV15 - String	ECNV14 - Not a string
date	format	ECV16 - Format is DD/MM/YYYY	ECNV15 - Format different from DD/MM/YYYY
date	day	ECV17 - Day is between 1 and 31	ECNV16 - Day less than 1
		BVV6 - Day 1	ECNV17 - Day more than 31
		BVV7 - Day 31	BVNV7 - Day 0
			BVNV8 - Day 32

date	month	ECV18 - Month is between 1 and 12	ECNV18 - Month less than 1
		BVV8 - Month 1	ECNV19 - Month more than 12
		BVV9 - Month 12	BVNV9 - Month 0
			BVNV10 - Month 13
date	year	ECV19 - Year is between 2025 and 2027	ECNV20 - Year before 2025
		BVV10 - Year 2025	ECNV21 - Year after 2027
		BVV11 - Year 2027	BVNV11- Year 2024
			BVNV12 - Year 2028
date	date valid	ECV20 - Date is equal or after the request date	ECNV22 - Date is before the request date
budget	datatype	ECV20 - Float	ECNV23 - Not a float
budget	decimal precision	ECV21 - Exactly 2 decimal positions	ECNV24 - Less than 2 decimal positions
		BVV12 - 2 decimals	ECNV25 - More than 2 decimal positions
			BVNV13 - 1 decimal
			BVNV14 - 3 decimals
budget	value	ECV22 - Between 50000.00 and 1000000.00	ECNV26 - Less than 50000.00
		BVV13 - 50000.00	ECNV27 - More than 1000000.00
		BVV14 - 1000000.00	BVNV15 - 49999.99
			BVNV16 - 1000000.01
output	datatype	ECV23 - String	ECNV28 - Not a string
output	format	ECV24 - MD5 format	ECNV29 - Not in MD5 format
output	size	ECV25 - 32 chars	ECNV30 - Less than 32 chars
		BVV15 - 32 chars	ECNV31 - More than 32 chars
			BVNV17 - 31 chars
			BVNV18 - 33 chars
output	save	ECV26 - Saved in indicated JSON file	ECNV32 - Saved somewhere different from indicated JSON file
output	duplicate	ECV27 - No exception	ECNV33 - Exception if project with the same name and CIF already exists
output	invalid	ECV28 - No exception	ECNV34 - Exception if any input data is invalid

These are the boundary values and equivalence classes we defined, they are quite self-explanatory.

Tests:

I/O	VALID/ INVALID	ID TEST	company_cif	project_acronym	DESCRIPTION	department	date	budget	RESULT	EC,BV COVERED
Input	VALID	Test1	"B98765431"	"ABCDE"	"10_char_de"	"HR"	1/1/2025	50000.00	Returns MD5 String (Project ID)	ECV1,2,3,4,5,6,7,8,9,10,11,15,16,17,18,19,20,21,22,23,24,25,26,27,28,BBVV1,2,4,6,8,10,13,15
Input	VALID	Test2	"A12345678"	"EMIGRATING"	"this_description_has_30_charas"	"FINANCE"	31/12/2027	10000.00	Returns MD5 String (Project ID)	ECV12 BVV 3,5,7,9,11,14
Input	VALID	Test3	"V8321393"	"CLOUDPYT"	"moving to cloud"	"LEGAL"	20/02/2026	1000000.00	Returns MD5 String (Project ID)	ECV13
Input	VALID	Test4	"34798256"	"CLOUDPYT"	"moving to cloud"	"LOGISTICS"	1/12/2025	50000.00	Returns MD5 String (Project ID)	ECV14
Input	INVALID	Test5	"12345678"	"123456789"	"moving to cloud"	"HR"	"20/02/2026"	70000.00	EnterpriseManagementException	ECNV2,, BVNV3
Input	INVALID	Test6	30000000	"CLOUDPYT"	"moving to cloud"	"LOGISTICS"	1/12/2025	50000.00	EnterpriseManagementException	ECNV1 BVNV1
Input	INVALID	Test7	"U35468256"	300	"moving to cloud"	"LOGISTICS"	2/12/2025	50000.01	EnterpriseManagementException	ECNV5 BVN3
Input	INVALID	Test8	"A58818501"	"CLOUDPYT"	"moving to cloud"	300	3/12/2025	50000.02	EnterpriseManagementException	ECNV12
Input	INVALID	Test9	"A58818501"	"CLOUDPYT"	"moving to cloud"	"MARKETING"	3/12/2025	50000.02	EnterpriseManagementException	ECNV13
Input	INVALID	Test10	"A58818501"	"CLOUDPYT"	"moving to cloud"	"LOGISTICS"	4/12/2025	50000.03	EnterpriseManagementException	ECNV14 BVN2
Input	INVALID	Test11	"A58818501"	"CLOUDPYT"	"moving to cloud"	"LOGISTICS"	"Jose Luis"	50000.03	EnterpriseManagementException	ECNV15 BVN2
Input	INVALID	Test12	"A58818501"	"CLOUDPYT11"	"moving to cloud"	"LOGISTICS"	5/12/2025	"5 potatoes"	EnterpriseManagementException	ECNV23 BVN4
Input	INVALID	Test13	"A58818501"	"CLOUDPYT"	"moving to cloud"	"LOGISTICS"	0/0/2023	50000.09	EnterpriseManagementException	ECNV20,22,18,16 , BVN9
Input	INVALID	Test14	"A58818501"	"CLOUDPYT"	"moving to cloud"	"LOGISTICS"	11/12/2025	60.000.123	EnterpriseManagementException	BVNV14
Input	INVALID	Test15	"A58818501"	"CLOUDPYT"	"moving to cloud"	"LOGISTICS"	11/12/2025	6000.00	EnterpriseManagementException	ECNV 26, BVN13,15,11
Input	INVALID	Test16	"A58818501"	"CLOUDPYT"	"moving to cloud"	"LOGISTICS"	16/12/2025	7000000.03	EnterpriseManagementException	ECNV24,27 BVNV12, 18
Input	INVALID	Test17	"A58818501"	"CLOUDPYT"	300	"LOGISTICS"	6/12/2025	50000.05	EnterpriseManagementException	ECNV9 BVN5
Input	INVALID	Test18	"A58818501"	"CLOUDPYT"	"hi"	"LOGISTICS"	01/12/2025	50000.06	EnterpriseManagementException	ECNV 10, BVN7
Input	INVALID	Test19	"A58818501"	"CLOUDPYT"	"this description is veryyyy long"	"LOGISTICS"	8/12/2025	50000.07	EnterpriseManagementException	ECNV 2,3 BVN6
Input	INVALID	Test20	"A58818501"	"TRISKADEIKAPHOBIA"	"moving to cloud"	"LOGISTICS"	31/12/2025	50000.08	EnterpriseManagementException	ECNV 6, BVN8
Input	INVALID	Test21	"A58818501"	"THIS"	"moving to cloud"	"LOGISTICS"	12/12/2025	50000.11	EnterpriseManagementException	ECNV 7, BVN10
Input	INVALID	Test22	"A58818501"	"CLOUDPYT"	"moving to cloud"	"LOGISTICS"	32/13/2028	50000.14	EnterpriseManagementException	ECNV 17,19,21 BVN8
Input	INVALID	Test23	"A58818501"	"@@@@@@"	"moving to cloud"	"LOGISTICS"	17/12/2025	50000.16	EnterpriseManagementException	ECNV 8
Output	INVALID	Test24	"A58818501"	"DUPTEST"	"different_desc"	"FINANCE"	02/01/2025	60000.00	EnterpriseManagementException	ECNV33
Output	INVALID	Test25	"A58818501"	"ABCDE"	"10_char_de"	"HR"	01/01/2025	50000.01	EnterpriseManagementException	ECNV34

For simplicity’s sake, we will divide the tests in groups:

Valid scenarios:

These tests prove that when the system is given perfectly valid data, it correctly registers the project, saves it to the JSON file, and returns the expected output.

- **Test 1:** Registers a valid HR project with a valid CIF. Verifies the project is successfully saved to the JSON file.
- **Test 2:** Registers a valid FINANCE project with a valid CIF. Verifies the project is successfully saved.
- **Test 3:** Registers a valid LEGAL project with a valid CIF. Verifies the project is successfully saved.
- **Test 4:** Registers a valid LOGISTICS project with a valid CIF. Verifies the project is successfully saved.
- **Test 25:** Tests the output of a valid registration to ensure the returned Project ID is strictly a String and exactly 32 characters long (Validating the MD5 Hash format).

Company CIF Validation

These tests ensure the system properly catches and rejects invalid CIF inputs.

- **Test 5:** Tests an invalid CIF format (for example, missing the control letter/number). Expects an "Invalid CIF format" exception.
- **Test 6:** Tests an invalid data type for the CIF (passing an integer instead of a string). Expects an "ERROR CIF must be a string" exception.

Project Acronym Validation

These tests verify that the project acronym strictly adheres to the length and character rules.

- **Test 7:** Tests an invalid data type for the acronym (passing an integer). Expects an exception.
- **Test 20:** Tests an acronym that is too short (less than 5 characters). Expects an exception.
- **Test 21:** Tests an acronym that is too long (more than 10 characters). Expects an exception.

- **Test 23:** Tests an acronym containing invalid symbols (@@@@@@). Ensures only alphanumeric characters are accepted.

Department Validation

These tests ensure the department field only accepts the predefined valid categories.

- **Test 8:** Tests an invalid data type for the department (passing an integer). Expects an exception.
- **Test 9:** Tests an unregistered department string (for example, "MARKETING"). Expects the system to reject it since it is not HR, FINANCE, LEGAL, or LOGISTICS.

Date Validation

These tests ensure the date string is formatted exactly as DD/MM/YYYY and is a real calendar date.

- **Test 10:** Tests an invalid data type for the date (passing an equation/integer instead of a string).
- **Test 11:** Tests an invalid date format (passing alphabetic characters like "Jose Luis").
- **Test 13:** Tests an impossible calendar date ("00/00/2023").
- **Test 22:** Tests out-of-bounds calendar days and months ("32/13/2028")

Budget Validation

These tests verify that the budget is a float, has correct decimal precision, and falls within the allowed financial limits.

- **Test 12:** Tests an invalid data type for the budget (passing a string instead of a float).
- **Test 14:** Tests invalid decimal precision (providing a float with more than 2 decimals). Expects an exception.
- **Test 15:** Tests a budget that is below the minimum allowed threshold (fails if budget < 50000.00).
- **Test 16:** Tests a budget that exceeds the maximum allowed threshold (fails if budget > 1000000.00).

Project Description Validation

These tests check that the description string falls within the allowed character lengths.

- **Test 17:** Tests an invalid data type for the description (passing an integer).
- **Test 18:** Tests a description that is too short (less than 10 characters). Expects an exception.
- **Test 19:** Tests a description that is too long (more than 30 characters). Expects an exception.

Business Logic (Duplicate Checking)

- **Test 24:** Tests the duplicate registration rule. Attempts to register a project using a CIF and Acronym combination that already exists in the system. Expects an "ERROR: A project with this name and CIF already exists" exception to prevent duplicate records.